CSA3714-SOFTWARE TESTING

LAB EXPERIMENTS

EXPERIMENT NO: 1

TOPIC: ATM SYSTEM – BUG IDENTIFICATION

OBJECTIVE

To identify and document software defects in an ATM system.

FEATURES TO BE TESTED

1. Card validation
2. User authentication
3. Account type selection
4. Cash withdrawal
5. Transaction and daily limits
6. Money denominations
7. Receipt printing
8. PIN change

BUG REPORT

Bug ID	**Bug Name**
ATM001	Invalid Card
ATM002	Invalid PIN
ATM003	Invalid Account Type
ATM004	Insufficient Balance
ATM005	Transaction Limit
ATM006	Daily Limit
ATM007	Invalid Money Denominations
ATM008	Receipt Not Printed
ATM009	PIN Change Mismatch
ATM010	Cash Not Dispensed

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TEST CASES

Test Case 1: Invalid Card

Test Case ID: ATM001

Steps: Insert an invalid ATM card.

Expected Result: The ATM should reject the card and display an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid PIN

Test Case ID: ATM002

Steps: Insert a valid card and enter an incorrect PIN.

Expected Result: An invalid PIN message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Invalid Account Type

Test Case ID: ATM003

Steps: Select an unsupported account type.

Expected Result: The ATM should reject the selection and display an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Insufficient Balance

Test Case ID: ATM004

Steps: Enter a withdrawal amount greater than the available balance.

Expected Result: Withdrawal should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Transaction Limit

Test Case ID: ATM005

Steps: Enter an amount exceeding the permitted transaction limit.

Expected Result: The ATM should display a transaction limit error.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Daily Limit

Test Case ID: ATM006

Steps: Attempt to withdraw beyond the daily withdrawal limit.

Expected Result: The transaction should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Invalid Money Denominations

Test Case ID: ATM007

Steps: Enter an amount that cannot be dispensed using the available notes.

Expected Result: The ATM should reject the amount and display a suitable message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Receipt Not Printed

Test Case ID: ATM008

Steps: Complete a withdrawal and request a receipt.

Expected Result: The ATM should print the receipt or display a printer error.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 9: PIN Change Mismatch

Test Case ID: ATM009

Steps: Enter different values in the new PIN and confirm PIN fields.

Expected Result: The ATM should reject the PIN change and display a mismatch message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 10: Cash Not Dispensed

Test Case ID: ATM010

Steps: Perform a valid cash withdrawal and check the cash dispenser and account balance.

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Expected Result: Cash should be dispensed, and the account should be debited correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, the ATM system was examined, and possible software defects were identified and documented.

EXPERIMENT NO: 2

TOPIC: ARMS PORTAL – POSITIVE TESTING

OBJECTIVE

To test the basic functions of the ARMS Portal using valid inputs.

FEATURES TO BE TESTED

1. User login
2. Student profile
3. Academic records
4. Student information update
5. Attendance details
6. Marks display
7. Logout

BUG REPORT / TEST CASE ID TABLE

Bug ID	**Bug Name**
ARMS001	Valid Login
ARMS002	Student Details Not Displayed
ARMS003	Academic Records Missing
ARMS004	Student Details Not Updated
ARMS005	Attendance Display Error
ARMS006	Marks Display Error
ARMS007	Logout Failure

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TEST CASES

Test Case 1: Valid Login

Test Case ID: ARMS001

Steps: Open the ARMS Portal, enter valid login credentials, and click Login.

Expected Result: The user should log in successfully, and the dashboard should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: View Student Details

Test Case ID: ARMS002

Steps: Log in and open the student profile.

Expected Result: Student details should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: View Academic Records

Test Case ID: ARMS003

Steps: Open the academic records section.

Expected Result: Academic records should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Update Student Details

Test Case ID: ARMS004

Steps: Enter valid information in an editable field and click Save.

Expected Result: The updated information should be saved successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: View Attendance

Test Case ID: ARMS005

Steps: Open the attendance section.

Expected Result: Attendance details should be displayed correctly.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: View Marks

Test Case ID: ARMS006

Steps: Open the marks section.

Expected Result: Marks should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Logout

Test Case ID: ARMS007

Steps: Log in and click the Logout button.

Expected Result: The user should be logged out and redirected to the login page.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of the ARMS Portal using valid inputs.

EXPERIMENT NO: 3

TOPIC: ARMS PORTAL – NEGATIVE TESTING

OBJECTIVE

To test the ARMS Portal using invalid inputs and verify its error-handling functions.

FEATURES TO BE TESTED

1. Login validation
2. Empty field validation
3. Student detail validation
4. Unauthorized access
5. Search validation
6. Session management
7. Logout

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BUG REPORT

Bug ID	Bug Name
ARMS-N001	Invalid Login Credentials
ARMS-N002	Empty Fields Accepted
ARMS-N003	Invalid Student Details Accepted
ARMS-N004	Unauthorized Access Allowed
ARMS-N005	Invalid Search Results
ARMS-N006	Session Timeout Error
ARMS-N007	Logout Failure

TEST CASES

Test Case 1: Invalid Login Credentials

Test Case ID: ARMS-N001

Steps: Enter an incorrect username or password and click Login.

Expected Result: Login should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Empty Fields Accepted

Test Case ID: ARMS-N002

Steps: Leave the username and password fields empty and click Login.

Expected Result: The portal should display validation messages.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Invalid Student Details Accepted

Test Case ID: ARMS-N003

Steps: Enter invalid data in an editable student information field and click Save.

Expected Result: Invalid data should be rejected with a suitable error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 4: Unauthorized Access Allowed

Test Case ID: ARMS-N004

Steps: Try to access a restricted page without logging in.

Expected Result: Access should be denied, and the login page should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Invalid Search Results

Test Case ID: ARMS-N005

Steps: Search using invalid or unavailable student details.

Expected Result: A suitable message should be displayed without showing incorrect records.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Session Timeout Error

Test Case ID: ARMS-N006

Steps: Leave the portal inactive until the configured session timeout period expires.

Expected Result: The session should expire securely, and the user should be asked to log in again.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Logout Failure

Test Case ID: ARMS-N007

Steps: Log in, click Logout, and try to access a protected page using the previous session.

Expected Result: The previous session should be terminated, and access should be denied.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test cases were prepared to verify whether the ARMS Portal handles invalid inputs and unauthorized access correctly.

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EXPERIMENT NO: 4

TOPIC: E-COMMERCE WEBSITE – POSITIVE TESTING

OBJECTIVE

To test the basic functions of an e-commerce website using valid inputs.

FEATURES TO BE TESTED

1. User registration
2. User login
3. Product search
4. Add to cart
5. Quantity update
6. Payment
7. Order placement

BUG REPORT / TEST CASE ID TABLE

Bug ID	**Bug Name**
EC001	User Registration Failure
EC002	Valid Login Failure
EC003	Product Search Error
EC004	Add to Cart Failure
EC005	Quantity Update Error
EC006	Payment Failure
EC007	Order Confirmation Missing

TEST CASES

Test Case 1: User Registration

Test Case ID: EC001

Steps: Enter valid registration details and click Register.

Expected Result: The account should be created successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 2: Valid Login

Test Case ID: EC002

Steps: Enter a registered username and correct password.

Expected Result: The user should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Product Search

Test Case ID: EC003

Steps: Search for an available product using its name.

Expected Result: Relevant products should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Add to Cart

Test Case ID: EC004

Steps: Select an available product and click Add to Cart.

Expected Result: The selected product should be added to the cart.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Quantity Update

Test Case ID: EC005

Steps: Increase the quantity of a product in the cart.

Expected Result: The quantity and total price should be updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Payment

Test Case ID: EC006

Steps: Complete the checkout using valid payment details.

Expected Result: Payment should be processed successfully, and the order should be recorded.

Actual Result: To be observed during testing.

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Status: Pass/Fail

Test Case 7: Order Confirmation

Test Case ID: EC007

Steps: Place an order successfully and view the order confirmation.

Expected Result: An order confirmation and order ID should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of an e-commerce website using valid inputs.

EXPERIMENT NO: 5

TOPIC: E-COMMERCE WEBSITE – NEGATIVE TESTING

OBJECTIVE

To test an e-commerce website using invalid inputs and verify its error-handling functions.

FEATURES TO BE TESTED

1. Registration validation
2. Login validation
3. Product availability
4. Quantity validation
5. Cart validation
6. Payment validation
7. Address validation
8. Order processing

BUG REPORT

Bug ID	**Bug Name**
EC-N001	Invalid Registration Details
EC-N002	Invalid Login Credentials
EC-N003	Out-of-Stock Product Accepted
EC-N004	Invalid Quantity Accepted

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EC-N005	Empty Cart Checkout
EC-N006	Invalid Payment Accepted
EC-N007	Invalid Address Accepted
EC-N008	Incorrect Total Price
EC-N009	Duplicate Order
EC-N010	Order Confirmation Missing

TEST CASES

Test Case 1: Invalid Registration Details

Test Case ID: EC-N001

Steps: Enter an invalid email address during registration and click Register.

Expected Result: Registration should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid Login Credentials

Test Case ID: EC-N002

Steps: Enter an incorrect username or password.

Expected Result: Login should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Out-of-Stock Product Accepted

Test Case ID: EC-N003

Steps: Try to add an out-of-stock product to the cart or place an order for it.

Expected Result: The website should prevent the purchase and display an out-of-stock message.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 4: Invalid Quantity Accepted

Test Case ID: EC-N004

Steps: Enter zero or a negative quantity for a product.

Expected Result: The website should reject the invalid quantity.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Empty Cart Checkout

Test Case ID: EC-N005

Steps: Try to proceed to checkout without adding any products to the cart.

Expected Result: Checkout should not be allowed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Invalid Payment Accepted

Test Case ID: EC-N006

Steps: Attempt payment using invalid payment details in a test environment.

Expected Result: Payment should be rejected, and an error message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Invalid Address Accepted

Test Case ID: EC-N007

Steps: Enter an incomplete delivery address and try to place the order.

Expected Result: The website should display validation messages and prevent order placement.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Incorrect Total Price

Test Case ID: EC-N008

Steps: Add products to the cart and compare the displayed total with the item prices and quantities.

Expected Result: The total price should be calculated correctly.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 9: Duplicate Order

Test Case ID: EC-N009

Steps: Submit the same order repeatedly or refresh the order submission page.

Expected Result: Duplicate orders should not be created unintentionally.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 10: Order Confirmation Missing

Test Case ID: EC-N010

Steps: Complete an order and check whether confirmation is displayed.

Expected Result: A confirmation message and order ID should be displayed for a successful order.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test cases were prepared to verify whether the e-commerce website handles invalid inputs, payment errors, and order-processing issues correctly.

EXPERIMENT NO: 6

TOPIC: FOOD DELIVERY APPLICATION – POSITIVE TESTING

OBJECTIVE

To identify and analyze the requirements of a Food Delivery Application and prepare positive test scenarios and test cases using valid inputs.

FEATURES TO BE TESTED

1. User registration and login.
2. Restaurant and food search.
3. View food details and prices.
4. Add food items to the cart.
5. Apply valid coupons.
6. Place orders and complete payment.

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7. Track orders and view order history.

BUG REPORT / TEST CASE TABLE

Bug ID	**Bug Name**
FDA001	User Registration Failure
FDA002	Valid Login Failure
FDA003	Food Search Error
FDA004	Add to Cart Failure
FDA005	Coupon Application Failure
FDA006	Order Placement Failure
FDA007	Payment Processing Failure
FDA008	Order Tracking Failure

TEST CASES

Test Case 1: User Registration

Test Case ID: FDA001

Steps:

1. Open the Food Delivery Application.
2. Enter valid user details.
3. Click Register.

Expected Result: The user account should be created successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Valid Login

Test Case ID: FDA002

Steps:

1. Enter the registered email or phone number.
2. Enter the correct password.
3. Click Login.

Expected Result: The user should log in successfully.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Food Search

Test Case ID: FDA003

Steps:

1. Open the application.
2. Search for an available food item.

Expected Result: Relevant food items should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Add to Cart

Test Case ID: FDA004

Steps:

1. Select an available food item.
2. Choose the quantity.
3. Click Add to Cart.

Expected Result: The selected item and quantity should appear in the cart.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Apply Coupon

Test Case ID: FDA005

Steps:

1. Add food items to the cart.
2. Enter a valid coupon code.
3. Click Apply.

Expected Result: The applicable discount should be calculated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 6: Place Order

Test Case ID: FDA006

Steps:

1. Add food items to the cart.
2. Select a delivery address.
3. Confirm the order.

Expected Result: The order should be placed successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Payment Processing

Test Case ID: FDA007

Steps:

1. Proceed to checkout.
2. Select a valid payment method.
3. Complete the payment.

Expected Result: Payment should be processed, and the order status should be updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Order Tracking

Test Case ID: FDA008

Steps:

1. Place an order successfully.
2. Open the order tracking section.

Expected Result: The current order status should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test scenarios and test cases were prepared to verify the basic functions of the Food Delivery Application using valid inputs.

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EXPERIMENT NO: 7

TOPIC: FOOD DELIVERY APPLICATION – NEGATIVE TESTING

OBJECTIVE

To analyze the requirements of a Food Delivery Application and prepare negative test scenarios and test cases to verify error handling.

FEATURES TO BE TESTED

1. Registration validation.
2. Login validation.
3. Food availability.
4. Cart quantity validation.
5. Coupon validation.
6. Delivery address validation.
7. Payment validation.
8. Duplicate order prevention.

BUG REPORT

Bug ID	**Bug Name**
FDA-N001	Invalid Registration Accepted
FDA-N002	Invalid Login Credentials Accepted
FDA-N003	Unavailable Food Item Accepted
FDA-N004	Invalid Quantity Accepted
FDA-N005	Invalid Coupon Accepted
FDA-N006	Invalid Address Accepted
FDA-N007	Failed Payment Treated as Successful
FDA-N008	Duplicate Order Created

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TEST CASES

Test Case 1: Invalid Registration

Test Case ID: FDA-N001

Steps:

1. Open the registration page.
2. Enter an invalid email address or leave a required field empty.
3. Click Register.

Expected Result: Registration should be rejected, and validation messages should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid Login

Test Case ID: FDA-N002

Steps:

1. Enter an incorrect password.
2. Click Login.

Expected Result: Login should be rejected with an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Unavailable Food Item

Test Case ID: FDA-N003

Steps:

1. Search for an unavailable food item.
2. Attempt to add it to the cart.

Expected Result: The application should indicate that the item is unavailable and prevent ordering it.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 4: Invalid Quantity

Test Case ID: FDA-N004

Steps:

1. Open the cart.
2. Enter zero or a negative quantity.

Expected Result: Invalid quantities should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Invalid Coupon

Test Case ID: FDA-N005

Steps:

1. Add items to the cart.
2. Enter an expired or invalid coupon code.
3. Click Apply.

Expected Result: The coupon should be rejected, and a suitable message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Invalid Address

Test Case ID: FDA-N006

Steps:

1. Proceed to checkout.
2. Enter an incomplete delivery address.
3. Attempt to place the order.

Expected Result: The application should request a valid delivery address.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 7: Failed Payment

Test Case ID: FDA-N007

Steps:

1. Proceed to checkout.
2. Simulate a failed payment using the test payment environment.

Expected Result: The payment should not be marked successful, and the order should not be confirmed as paid.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Duplicate Order

Test Case ID: FDA-N008

Steps:

1. Submit an order.
2. Repeatedly click the Place Order button or refresh the submission page.

Expected Result: Only one order should be created for the same submission.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test scenarios and test cases were prepared to verify the Food Delivery Application's ability to handle invalid inputs, payment failures, and duplicate orders.

EXPERIMENT NO: 8

TOPIC: BANKING APPLICATION – POSITIVE TESTING

OBJECTIVE

To identify and analyze the requirements of a Banking Application and prepare positive test scenarios and test cases using valid inputs.

FEATURES TO BE TESTED

1. Customer registration and login.
2. Account balance enquiry.
3. Fund transfer.

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4. Money deposit.
5. Money withdrawal.
6. Transaction history.
7. Bill payment.
8. Logout.

BUG REPORT / TEST CASE TABLE

Bug ID	**Bug Name**
BANK001	Customer Registration Failure
BANK002	Valid Login Failure
BANK003	Balance Display Error
BANK004	Fund Transfer Failure
BANK005	Deposit Processing Failure
BANK006	Withdrawal Processing Failure
BANK007	Transaction History Error
BANK008	Bill Payment Failure

TEST CASES

Test Case 1: Customer Registration

Test Case ID: BANK001

Steps: Enter valid customer details and submit the registration form.

Expected Result: The customer should be registered successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Valid Login

Test Case ID: BANK002

Steps: Enter the registered username, correct password, and required authentication details.

Expected Result: The customer should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 3: Balance Enquiry

Test Case ID: BANK003

Steps: Log in and open the account balance section.

Expected Result: The correct available balance should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Fund Transfer

Test Case ID: BANK004

Steps: Enter a valid beneficiary account and transfer amount within the available balance.

Expected Result: The transfer should be completed, and both account records should be updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Deposit Processing

Test Case ID: BANK005

Steps: Perform a valid deposit through the supported banking channel.

Expected Result: The deposited amount should be credited correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Withdrawal Processing

Test Case ID: BANK006

Steps: Request a withdrawal within the available balance and permitted limit.

Expected Result: The withdrawal should succeed, and the balance should be updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Transaction History

Test Case ID: BANK007

Steps: Open the transaction history section.

Expected Result: Previous transactions should be displayed with correct dates and amounts.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Bill Payment

Test Case ID: BANK008

Steps: Select a supported biller, enter valid details, and complete payment.

Expected Result: The payment should be processed and a confirmation displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test scenarios and test cases were prepared to verify the basic functions of the Banking Application using valid inputs.

EXPERIMENT NO: 9

TOPIC: BANKING APPLICATION – NEGATIVE TESTING

OBJECTIVE

To analyze the requirements of a Banking Application and prepare negative test scenarios and test cases to verify its error handling and security.

FEATURES TO BE TESTED

1. Login validation.
2. Account access control.
3. Balance validation.
4. Fund transfer validation.
5. Withdrawal limits.
6. Beneficiary validation.
7. Payment processing.
8. Session management.

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BUG REPORT

Bug ID	Bug Name
BANK-N001	Invalid Login Accepted
BANK-N002	Unauthorized Account Access
BANK-N003	Incorrect Balance Display
BANK-N004	Invalid Fund Transfer Accepted
BANK-N005	Excess Withdrawal Accepted
BANK-N006	Invalid Beneficiary Accepted
BANK-N007	Failed Payment Marked Successful
BANK-N008	Session Not Terminated

TEST CASES

Test Case 1: Invalid Login

Test Case ID: BANK-N001

Steps: Enter an incorrect password and click Login.

Expected Result: Login should be rejected with an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Unauthorized Account Access

Test Case ID: BANK-N002

Steps: Attempt to access another customer's account without permission.

Expected Result: Access should be denied.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Incorrect Balance Display

Test Case ID: BANK-N003

Steps: Compare the displayed balance with the verified account balance after a test transaction.

Expected Result: The displayed balance should match the correct account records.

Actual Result: To be observed during testing.

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Status: Pass/Fail

Test Case 4: Invalid Fund Transfer

Test Case ID: BANK-N004

Steps: Attempt to transfer a negative amount or an amount exceeding the available balance.

Expected Result: The transfer should be rejected, and account balances should remain correct.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Excess Withdrawal

Test Case ID: BANK-N005

Steps: Request a withdrawal exceeding the available balance or permitted withdrawal limit.

Expected Result: The withdrawal should be rejected.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Invalid Beneficiary

Test Case ID: BANK-N006

Steps: Enter an invalid beneficiary account number and attempt a transfer.

Expected Result: The application should reject the transfer and display an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Failed Payment

Test Case ID: BANK-N007

Steps: Simulate a failed bill payment using an approved test environment.

Expected Result: The payment should not be marked successful, and the account should not be incorrectly debited.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Session Not Terminated

Test Case ID: BANK-N008

Steps: Log out and attempt to reopen a protected account page using the previous session.

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Expected Result: Access should be denied, and reauthentication should be required.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test scenarios and test cases were prepared to verify the Banking Application's handling of invalid transactions, unauthorized access, and security errors.

EXPERIMENT NO: 10

TOPIC: TEST PLAN FOR A LIBRARY MANAGEMENT SYSTEM

OBJECTIVE

To develop a comprehensive Test Plan for a Library Management System to verify its functionality, security, usability, and performance.

FEATURES TO BE TESTED

1. User login and authentication
2. Book registration and management
3. Book search
4. Book issue and return
5. Due date and fine calculation
6. Book reservation
7. Student record management
8. Report generation

TEST PLAN

1. Scope of Testing

- Verify user login and access permissions.
- Test adding, updating, searching, and deleting book records.
- Verify book issue and return operations.
- Test due dates and fine calculations.
- Verify book reservations and availability.
- Check report generation.

2. Testing Strategies

- Functional Testing: Verify all system functions.

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- Integration Testing: Check communication between modules.
- System Testing: Test the complete application.
- Security Testing: Verify login and access permissions.
- Performance Testing: Check system response under multiple users.
- Regression Testing: Ensure changes do not affect existing functions.

3. Test Design Techniques

- Equivalence Partitioning: Divide inputs into valid and invalid groups.
- Boundary Value Analysis: Test values near borrowing limits.
- Decision Table Testing: Check borrowing conditions.
- State Transition Testing: Verify book states such as Available, Issued, and Returned.
- Error Guessing: Test common mistakes such as missing book details.

4. Test Execution Strategy

1. Analyze the requirements.
2. Prepare test scenarios and test cases.
3. Set up the test environment.
4. Execute test cases.
5. Record actual results.
6. Report and retest defects.
7. Perform regression testing.
8. Prepare the final test report.

5. Entry Criteria

- Requirements are documented.
- The application is available.
- Test cases and test data are prepared.
- The test environment is ready.

6. Exit Criteria

- All planned test cases have been executed.
- Critical defects are resolved or formally accepted.
- Failed test cases are retested.

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- Regression testing is completed.
- Test results are documented.

BUG REPORT

Bug ID	**Bug Name**
LMS001	Login Failure
LMS002	Book Search Error
LMS003	Book Details Not Saved
LMS004	Book Issue Failure
LMS005	Book Return Failure
LMS006	Incorrect Fine Calculation
LMS007	Reservation Failure
LMS008	Report Generation Error

TEST CASES

Test Case 1: Login Failure

Test Case ID: LMS001

Steps: Enter valid login credentials and click Login.

Expected Result: The user should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Book Search Error

Test Case ID: LMS002

Steps: Search for an available book using its title.

Expected Result: Correct book details should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Book Details Not Saved

Test Case ID: LMS003

Steps: Enter valid book details and click Save.

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Expected Result: Book details should be saved successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Book Issue Failure

Test Case ID: LMS004

Steps: Issue an available book to an eligible student.

Expected Result: The book should be issued successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Book Return Failure

Test Case ID: LMS005

Steps: Return an issued book and update its status.

Expected Result: The book status should change to Available, subject to reservation rules.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Incorrect Fine Calculation

Test Case ID: LMS006

Steps: Return a book after its due date and check the fine.

Expected Result: The fine should be calculated according to the configured rules.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Reservation Failure

Test Case ID: LMS007

Steps: Reserve an eligible book that is currently unavailable.

Expected Result: The reservation should be recorded successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 8: Report Generation Error

Test Case ID: LMS008

Steps: Open the reports section and generate a book issue report.

Expected Result: The report should display accurate information.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, a comprehensive Test Plan was prepared to evaluate the functionality, security, usability, and performance of the Library Management System.

EXPERIMENT NO: 11

TOPIC: RAILWAY RESERVATION SYSTEM – POSITIVE TESTING

OBJECTIVE

To identify and analyze the requirements of a Railway Reservation System and prepare positive test scenarios and test cases using valid inputs.

FEATURES TO BE TESTED

1. User registration
2. User login
3. Train search
4. Seat availability
5. Passenger details
6. Ticket booking
7. Payment processing
8. PNR status checking

BUG REPORT

Bug ID	**Bug Name**
RRS001	User Registration Failure
RRS002	Valid Login Failure
RRS003	Train Search Error
RRS004	Seat Availability Error

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RRS005	Passenger Details Not Saved
RRS006	Ticket Booking Failure
RRS007	Payment Processing Failure
RRS008	PNR Status Error

TEST CASES

Test Case 1: User Registration

Test Case ID: RRS001

Steps: Enter valid user details and submit the registration form.

Expected Result: The user account should be created successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Valid Login

Test Case ID: RRS002

Steps: Enter valid login credentials and click Login.

Expected Result: The user should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Train Search

Test Case ID: RRS003

Steps: Enter a valid source, destination, and journey date.

Expected Result: Available trains should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Seat Availability

Test Case ID: RRS004

Steps: Select a train and check seat availability.

Expected Result: Correct seat availability should be displayed.

Actual Result: To be observed during testing.

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Status: Pass/Fail

Test Case 5: Passenger Details

Test Case ID: RRS005

Steps: Enter valid passenger details and continue.

Expected Result: Passenger details should be saved correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Ticket Booking

Test Case ID: RRS006

Steps: Select an available train, enter passenger details, and confirm the booking.

Expected Result: The booking should be created successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Payment Processing

Test Case ID: RRS007

Steps: Complete the booking using a valid payment method.

Expected Result: Payment should be processed and booking status updated correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: PNR Status

Test Case ID: RRS008

Steps: Enter a valid PNR number.

Expected Result: The correct ticket status should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of the Railway Reservation System using valid inputs.

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EXPERIMENT NO: 12

TOPIC: RAILWAY RESERVATION SYSTEM – NEGATIVE TESTING

OBJECTIVE

To identify and analyze the requirements of a Railway Reservation System and prepare negative test scenarios and test cases using invalid inputs.

FEATURES TO BE TESTED

1. Login validation
2. Train search validation
3. Journey date validation
4. Seat availability validation
5. Passenger detail validation
6. Duplicate booking prevention
7. Payment failure handling
8. PNR validation

BUG REPORT

Bug ID	**Bug Name**
RRS-N001	Invalid Login Accepted
RRS-N002	Invalid Train Search Accepted
RRS-N003	Past Journey Date Accepted
RRS-N004	Unavailable Seat Booking Allowed
RRS-N005	Invalid Passenger Details Accepted
RRS-N006	Duplicate Booking Created
RRS-N007	Failed Payment Marked Successful
RRS-N008	Invalid PNR Accepted

TEST CASES

Test Case 1: Invalid Login

Test Case ID: RRS-N001

Steps: Enter an incorrect password and click Login.

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Expected Result: Login should be rejected with an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Invalid Train Search

Test Case ID: RRS-N002

Steps: Enter an invalid source or destination station.

Expected Result: The system should reject invalid station details.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Past Journey Date

Test Case ID: RRS-N003

Steps: Select a journey date in the past.

Expected Result: The system should prevent booking for an invalid past journey date.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Unavailable Seat Booking

Test Case ID: RRS-N004

Steps: Attempt to book a confirmed seat when no seats are available.

Expected Result: The system should prevent the unavailable confirmed booking or offer an eligible waiting-list option.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Invalid Passenger Details

Test Case ID: RRS-N005

Steps: Leave required passenger fields empty or enter invalid data.

Expected Result: Validation messages should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 6: Duplicate Booking

Test Case ID: RRS-N006

Steps: Submit the same booking request repeatedly.

Expected Result: Duplicate bookings should not be created unintentionally.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Failed Payment

Test Case ID: RRS-N007

Steps: Simulate a failed payment in the test environment.

Expected Result: Payment should not be marked successful, and booking status should be handled correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Invalid PNR

Test Case ID: RRS-N008

Steps: Enter an invalid or nonexistent PNR number.

Expected Result: An invalid PNR message should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, negative test cases were prepared to verify the Railway Reservation System's handling of invalid inputs, unavailable seats, and payment errors.

EXPERIMENT NO: 13

TOPIC: ONLINE EXAMINATION SYSTEM – POSITIVE TESTING

OBJECTIVE

To identify and analyze the requirements of an Online Examination System and prepare positive test scenarios and test cases using valid inputs.

FEATURES TO BE TESTED

1. Student login
2. Examination schedule

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3. Instructions display
4. Question display
5. Answer selection
6. Timer functionality
7. Exam submission
8. Result generation

BUG REPORT

Bug ID	**Bug Name**
OES001	Valid Login Failure
OES002	Exam Schedule Not Displayed
OES003	Instructions Not Displayed
OES004	Questions Not Displayed
OES005	Answers Not Saved
OES006	Timer Display Error
OES007	Exam Submission Failure
OES008	Result Generation Error

TEST CASES

Test Case 1: Valid Login

Test Case ID: OES001

Steps: Enter valid student login credentials and click Login.

Expected Result: The student should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Exam Schedule

Test Case ID: OES002

Steps: Open the examination schedule.

Expected Result: The correct examination date, time, and subject should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

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Test Case 3: Instructions Display

Test Case ID: OES003

Steps: Open the scheduled examination and view the instructions.

Expected Result: Examination instructions should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Question Display

Test Case ID: OES004

Steps: Start an authorized examination.

Expected Result: Questions and answer options should be displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Answers Not Saved

Test Case ID: OES005

Steps: Select answers and move between questions.

Expected Result: Selected answers should be saved correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Timer Functionality

Test Case ID: OES006

Steps: Start an examination and observe the countdown timer.

Expected Result: The timer should display the remaining time correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Exam Submission

Test Case ID: OES007

Steps: Answer the questions and click Submit Exam.

Expected Result: The examination should be submitted successfully.

Actual Result: To be observed during testing.

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Status: Pass/Fail

Test Case 8: Result Generation

Test Case ID: OES008

Steps: Complete the examination and open the result section when results are available.

Expected Result: The correct examination result should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, positive test cases were prepared to verify the basic functions of the Online Examination System using valid inputs.

EXPERIMENT NO: 14

TOPIC: ONLINE EXAMINATION SYSTEM – NEGATIVE TESTING

OBJECTIVE

To identify and analyze the requirements of an Online Examination System and prepare negative test scenarios and test cases to verify error handling and security.

FEATURES TO BE TESTED

1. Login validation
2. Examination access control
3. Schedule validation
4. Answer validation
5. Timer handling
6. Submission validation
7. Duplicate submission prevention
8. Result access control

BUG REPORT

Bug ID	**Bug Name**
OES-N001	Invalid Login Accepted
OES-N002	Unauthorized Exam Access
OES-N003	Exam Started Outside Schedule

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OES-N004	Invalid Answer Accepted
OES-N005	Timer Failure
OES-N006	Empty Submission Handling Error
OES-N007	Duplicate Submission Accepted
OES-N008	Unauthorized Result Access

TEST CASES

Test Case 1: Invalid Login

Test Case ID: OES-N001

Steps: Enter an incorrect username or password.

Expected Result: Login should be rejected with an error message.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Unauthorized Exam Access

Test Case ID: OES-N002

Steps: Attempt to open an examination without valid authentication or permission.

Expected Result: Access should be denied.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Exam Started Outside Schedule

Test Case ID: OES-N003

Steps: Attempt to start an examination before its permitted start time or after its closing time.

Expected Result: The system should prevent unauthorized examination access.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Invalid Answer

Test Case ID: OES-N004

Steps: Submit an invalid answer value in the controlled test environment.

Expected Result: Invalid answer data should be rejected.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Timer Failure

Test Case ID: OES-N005

Steps: Start a test examination and allow the timer to reach zero.

Expected Result: The examination should follow the configured time-expiry and submission rules.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Empty Submission

Test Case ID: OES-N006

Steps: Attempt to submit the examination without answering any questions.

Expected Result: The system should follow its configured submission rules and indicate unanswered questions when applicable.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Duplicate Submission

Test Case ID: OES-N007

Steps: Click Submit repeatedly or refresh the page after submitting.

Expected Result: Multiple unintended submissions should not be recorded.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Unauthorized Result Access

Test Case ID: OES-N008

Steps: Attempt to access another student's result without permission.

Expected Result: Access should be denied.

Actual Result: To be observed during testing.

Status: Pass/Fail

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RESULT

Thus, negative test cases were prepared to verify the Online Examination System's handling of invalid inputs, unauthorized access, and submission errors.

EXPERIMENT NO: 15

TOPIC: TEST PLAN FOR AN INVENTORY MANAGEMENT SYSTEM

OBJECTIVE

To develop a comprehensive Test Plan for an Inventory Management System covering test scope, test design techniques, test execution strategy, and test completion criteria.

FEATURES TO BE TESTED

1. User login and access control
2. Product registration and management
3. Stock addition and removal
4. Stock level updates
5. Supplier management
6. Purchase and sales records
7. Low-stock alerts
8. Inventory reports

TEST PLAN

1. Scope of Testing

- Verify login and user permissions.
- Test adding, updating, searching, and deleting products.
- Verify stock addition and removal.
- Test purchase and sales transactions.
- Verify stock quantity calculations.
- Test low-stock alerts.
- Verify supplier information.
- Test inventory report generation.

2. Testing Strategies

- Functional Testing: Verify all inventory functions.

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- Integration Testing: Check communication between product, supplier, purchase, and sales modules.
- System Testing: Test the complete application.
- Security Testing: Verify authentication and access permissions.
- Performance Testing: Check system response under multiple transactions.
- Usability Testing: Ensure the application is easy to use.
- Regression Testing: Ensure updates do not affect existing functions.

3. Test Design Techniques

- Equivalence Partitioning: Divide stock quantities into valid and invalid groups.
- Boundary Value Analysis: Test values around minimum and maximum stock limits.
- Decision Table Testing: Check combinations of stock availability, order quantity, and user permissions.
- State Transition Testing: Verify states such as In Stock, Low Stock, and Out of Stock.
- Error Guessing: Test common mistakes such as duplicate product IDs and negative quantities.

4. Test Execution Strategy

1. Analyze requirements.
2. Prepare test scenarios and test cases.
3. Set up the test environment.
4. Execute test cases.
5. Record actual results and compare them with expected results.
6. Report defects and assign severity.
7. Retest corrected defects.
8. Perform regression testing.
9. Prepare the final test report.

5. Entry Criteria

- Requirements are documented.
- The application is available for testing.
- Test cases and test data are prepared.
- The test environment is ready.

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6. Test Completion Criteria

- All planned test cases have been executed.
- Critical defects are resolved or formally accepted.
- Failed test cases have been investigated and retested.
- Stock calculations and transaction records are verified.
- Regression testing is completed.
- Test results and outstanding defects are documented.

7. BUG REPORT

Bug ID	**Bug Name**
IMS001	Login Failure
IMS002	Product Details Not Saved
IMS003	Incorrect Stock Quantity
IMS004	Stock Addition Failure
IMS005	Stock Removal Failure
IMS006	Supplier Details Error
IMS007	Low-Stock Alert Failure
IMS008	Inventory Report Error

TEST CASES

Test Case 1: Login Failure

Test Case ID: IMS001

Steps: Enter valid login credentials and click Login.

Expected Result: The user should log in successfully.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 2: Product Details Not Saved

Test Case ID: IMS002

Steps: Enter valid product details and click Save.

Expected Result: Product details should be saved successfully.

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Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 3: Incorrect Stock Quantity

Test Case ID: IMS003

Steps: Add or remove stock and verify the displayed quantity.

Expected Result: The stock quantity should be updated accurately.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 4: Stock Addition Failure

Test Case ID: IMS004

Steps: Enter a valid product and quantity to add stock.

Expected Result: The stock quantity should increase correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 5: Stock Removal Failure

Test Case ID: IMS005

Steps: Remove a valid quantity of available stock.

Expected Result: The stock quantity should decrease correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 6: Supplier Details Error

Test Case ID: IMS006

Steps: Add valid supplier details and save the record.

Expected Result: Supplier information should be stored and displayed correctly.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 7: Low-Stock Alert Failure

Test Case ID: IMS007

Steps: Reduce the stock quantity below the configured minimum level.

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Expected Result: A low-stock alert should be displayed.

Actual Result: To be observed during testing.

Status: Pass/Fail

Test Case 8: Inventory Report Error

Test Case ID: IMS008

Steps: Generate an inventory report.

Expected Result: The report should display correct product and stock information.

Actual Result: To be observed during testing.

Status: Pass/Fail

RESULT

Thus, a comprehensive Test Plan was prepared to evaluate the Inventory Management System's functionality, stock accuracy, security, and reliability.

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